Robin Cserveny

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Research Interests

My research lies at the intersection of theoretical atomic physics, quantum science, and computational physics. I focus on heavy ions, which are crucial for testing fundamental physical theories, including investigating physics beyond the Standard Model and searching for Dark Matter signatures. For my master's thesis, I performed high-precision atomic structure calculations for single-valence heavy ions, directly contributing to the development of next-generation optical atomic clocks. I am eager to extend this research to multi-valence systems for my doctoral project, pushing the boundaries of precision measurements by supporting and proposing new atomic clock candidates to address open questions in fundamental physics.

Education

PhD in Theoretical Physics 07/2026 – present
University of Queensland (UQ) Brisbane, Australia

PhD topic: Search for new physics beyond the Standard Model

MSc in Physics – GPA: 1.28 10/2022 – 02/2025
University of Vienna (UniVie) Vienna, Austria

Exchanges: University of Florida (US) (Spring Term 2024) – GPA: 4.0/4.0
University of Queensland (AU) (Full Year 2023) – GPA: 6.67/7.0

Honours: Graduated with honours (UniVie, 2025)

Awards: Dean's List Award in recognition for academic excellence (UniVie, 2025)
Dean's Commendation for Academic Excellence (UQ, 2023)

Thesis: Computational characterisation of Ra^+ and Sr^+ for the development of optical atomic clocks

- Achieved best possible grade – excellent (1)
- Presented results at ANZCOP–AIP Summer Meeting 2023
- Published results in Physical Review A as 1st author

BSc in Technical Physics – GPA: 1.7 10/2018 – 07/2022
TU Wien Vienna, Austria

Exchange: Loughborough University (UK) (Spring Term 2022) – First Class Honours (best possible grade)

Thesis: Sputter Yield Measurements of 2keV Ar^+ and $D_2{}^+$ and Estimation of Adsorbate Layer Thicknesses

- Achieved best possible grade – excellent (1)
- Earned co-authorship in 2 publications

BG/BRG Keimgasse Mödling 2012 – 2016
Oberstufe Mödling, Austria

- Was taught in the so-called "Modelklasse" system
- Completed all minor subjects within 2 years
- Took "Pluskurse" in physics, biology, and chemistry in addition

Sportgymnasium Maria Enzersdorf 2008 – 2012
Unterstufe Maria Enzersdorf, Austria

- Chose the option „Englisch als Arbeitssprache"
- Various subjects taught in English on a regular basis

Research Experience

Theoretical characterisation of systems for the development of atomic clocks 08/2023 – 08/2024
Atomic Theory Group (UQ) Brisbane, AU

- Investigated single-valence ionic systems for potential use as optical atomic clocks (OACs)
- Performed high-precision atomic structure calculations on HPC cluster
- Quantified systematic effects during operation of OACs
- Calculated and published previously unavailable key quantities

Sputtering of first-wall materials in fusion reactors 12/2020 – 06/2021
Atomic and Plasma Physics Group (TU Wien) Vienna, Austria

- Performed laboratory simulations of plasma-induced sputtering effects in fusion reactors
- Integrated and automated a new control module, enabling rapid switching of ion species
- Improved control of exp. parameters and reproducibility of experiments
- Developed code in Python that is still in use

Relevant professional Experience

Software Developer 02/2025 – present
Papyrus Software Brunn am Gebirge, Austria

- Analysed, developed and tested pattern-search algorithms in C++
- Developed software for document capture and processing systems
- Integrated cloud-based AI services (MS Azure, AWS) via REST API
- Used Query Language (SQL) to define data extraction workflows

Civilian Service Worker 01/2017 – 09/2017
SOS Kinderdorf Hinterbuehl Hinterbrühl, Austria

- Took care of children, most being diagnosed with psychiatric disorders
- Assisted with daily tasks: cooking, home work, medical appointments etc.
- Completed de-escalation training (PART) to handle conflicts

Teaching Experience

University Teaching Assistant 03/2020 – 02/2023
Institute of Analysis and Scientific Computing, TU Wien Vienna, Austria

- Led problem classes in Practical Analysis I for Technical Physics students
- Taught introductory mathematics (50+ participants)
- Explained key mathematical concepts and problem-solving strategies
- Created and verified solutions to problem sheets prior to publishing

Awards[1]

Dean's List Award in recognition of academic excellence 2025
University of Vienna

Graduated with honours 2025
University of Vienna

Dean's Commendation for Academic Excellence 2023
University of Queensland

[1] Hyperlinks are provided by hovering over the respective sections

Publications[1]

R. Cserveny, B. Roberts. Theoretical characterization of the barium II and radium II ions.
Physical Review A. 2025, 112, 032816. doi: 10.1103/1rtb-8ymc
Journal metrics: JIF: 2.6 – JCI: 0.83 – CiteScore: 5.3

I presented theoretical results of my Master's thesis in this work, for which I independently carried out all the computational work, created the presented visualisations and tables, and drafted the manuscript.

C. Cupak, F. Brandstätter, **R. Cserveny** et al. Absence of synergistic effects in quasi-simultaneous sputtering of tungsten by Ar and D ions. Nuclear Materials and Energy. 2023, 23, 101435. doi: 10.1016/j.nme.2023.10143
Journal metrics: JIF: 2.3 – JCI: 1.11 – CiteScore: 3.7

The results from my bachelor's thesis formed the foundation of this paper. I developed and optimised the software used to carry out the experiments to obtain the data for publication. Additionally, I was involved in the writing process by thoroughly proofreading the manuscript prior to publication.

Conferences[1]

R. Cserveny, B. Roberts. ANZCOP-AIP Summer Meeting 2023, The Australian National University, Canberra, ACT, Australia. Talk: Theoretical characterisation of the Ra ion for the development of atomic clocks and studies of fundamental physics.

I presented data collected over the course of my master's thesis, for which I performed all calculations and created detailed visualisations. I took full responsibility for preparing and delivering the presentation.

C. Cupak, R. Stadlmayr, H. Biber, P. Szabo, **R. Cserveny**, M. Fellinger, F. Aumayr (2021).
Studying plasma-wall-interaction processed in the laboratory using a sensitive quartz crystal microbalance.
Proc. of the XXV International Conference on Ion-Surface Interactions (ISI-2021).

I earned co-authorship by developing software that was integral to the experiments and by contributing experimental data generated during my bachelor's thesis.

Technical Skills and Certificates[1]

Programming: C++, Python, SQL

Libraries: OpenMP, NumPy, Matplotlib, scikit-learn, pandas

Software and tools: Git, Gerrit, MS Visual Studio, UNIX (terminal), LaTeX

VSC School[2] certificates: Modern C++ Software Design and Python for High Performance Computing

Other certificates: Mental Health First Aid Training
PART Training: training program for prevention and de-escalation of conflicts
PADI Advanced Open Water Diver

Languages[1]

German: Native

English: C2 — IELTS Academic - Overall Band Score 8.0

Spanish: B1 — SIELE

French: B1 — DELF

[1] Hyperlinks are provided by hovering over the respective sections
[2] Vienna Scientific Cluster